

CONNOR YABLONSKI

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MS in Data Science student at USF with experience in astronomy research and Retrieval Augmented Generation (RAG).

Education

University of San Francisco

M.S. in Data Science

San Francisco, CA

July 2025 - July 2026

University of Washington

B.S. in Physics and Astronomy

Seattle, WA

September 2020 - June 2024

Experience

Data Scientist

Gaia-AI

San Francisco, CA

October 2025 - Present

- Developed multi-step and recursive LSTM forecasting models in PyTorch with integrated MLflow for tracking and evaluation
- Engineered road, water, and land-based geospatial features using GIS transformations, k-means anchor point encoding, and H3 spatial tiling, reducing mean distance error by approximately 30%
- Maintained and enhanced a end-to-end machine learning pipeline supporting XGBoost and Random Forest models by automating EDA, model evaluation, and visualization workflows

Data Scientist

Vera C. Rubin Observatory, Research Aide

Seattle, WA

February 2025 - June 2025

- Designed and implemented a Retrieval-Augmented Generation (RAG) chatbot from Rubin Observatory documentation, leading ingestion, embedding, and data pipeline development using LangChain and a Weaviate vector database.
- Independently developed scraping and ingestion pipelines to extract, structure, and embed documentation from 8 distinct data sources into a semantic search system.
- Deployed a Streamlit web app using Kubernetes, Helm, and ArgoCD, resulting in a chatbot that allowed scientists to quickly query from over 20 years of documentation.

Projects

Advanced ML

University of San Francisco

San Francisco, CA

February 2026 - March 2026

- Developed PyTorch LSTM/MLP and scikit-learn Random Forest models for many-to-one sequential regression, integrating MLflow for hyperparameter tracking and comparing PCA against aggregation statistics for dimensionality reduction.

ML Ops: Hikeability

University of San Francisco

San Francisco, CA

March 2026 - May 2026

- Developed hikeability, an end-to-end LLM trail classification platform leveraging MLflow for prompt tuning, automated GitHub Actions CI/CD pipelines for daily scraping, and a Vercel-hosted Flask backend with a dynamic Mapbox frontend.

Experimentation

University of San Francisco

San Francisco, CA

January 2026

- Led a multi-stage A/B testing initiative using factorial design and causal inference to optimize UI interactions on a simulated Netflix homepage, reducing average browsing time by 2 minutes and 20 seconds within a strict 10,000-unit budget.

Distributed Data Systems

University of San Francisco

San Francisco, CA

February 2026 - March 2026

- Orchestrated an automated Apache Airflow pipeline integrating GCS, MongoDB, and BigQuery ML to ingest daily weather observations and train a linear regression model predicting next-day maximum temperatures with an MAE of 3.86°F.

Service

Executive Committee

Seattle, WA

University of Washington, Kappa Sigma

April 2022 - April 2023

- Elected to the top leadership position as an Executive Committee member, where I guided the chapter to win 5 awards (including recognition as one of the top 3% of chapters nationwide), led the Public Relations Committee and Housing Committee, and drove a 33% increase in membership

Publications

- Photometry Paper: <https://arxiv.org/pdf/2502.03179>
- Upcoming Paper & SPIE Presentation: Development of a Retrieval-Augmented Generation Chatbot for Enhanced Information Discovery at Rubin Observatory

Technical Skills

Languages: Python (Pandas, NumPy, Matplotlib, BeautifulSoup, LangChain), SQL, LaTeX, Bash

Tools: Deployment (Docker, FastAPI, Helm, Kubernetes, ArgoCD), ML (XGBoost, RandomForest, LSTM, Transformers, scikit-learn, MLFlow, PyTorch, TensorFlow, Keras), Data Processing (Postgres, Apache Spark, Spark SQL, MongoDB, Airflow, GCP, BigQuery), AI (Weaviate vector database, LangChain, RAG), Version Control (Git, GitHub), EDA (Matplotlib, Seaborn, Plotly, Tableau), GIS, Microsoft Office